PHYSICS · CLASS 11 · JEE

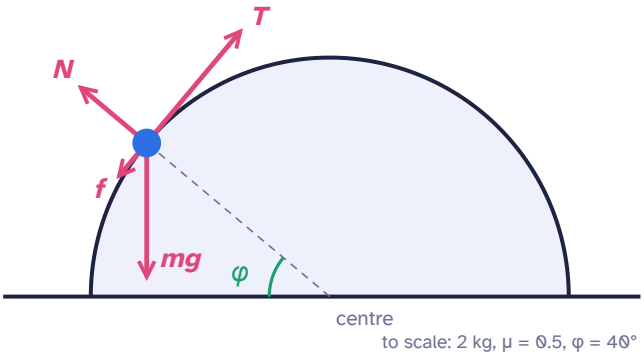
Work done dragging a block over a rough dome

Class notes · step by step, from zero · matches the video lesson

The question. A hemisphere of radius R is fixed on the ground. Its surface is rough, with coefficient of friction μ. Using a string, a person pulls a small block of mass m very slowly along the surface, from the ground all the way up to the top of the hemisphere. Find the work done by the person.

What we know

- Radius of the dome: R
- Mass of the block: m
- Friction coefficient: μ
- The block moves **slowly**
- Find: W_{person}



The plan

1. Draw every force on the block.
2. **Slowly** ⇒ forces balance ⇒ the pull T.
3. T keeps changing, so add up tiny pieces: $W = \int T\, ds$.
4. Check the answer with energy.

1. What is work?

A force does work when it moves something. If the force is at angle θ to the motion, only the part along the motion counts:

$$W = F\, s \cos\theta$$

Here the pull changes as the block climbs, so we cut the path into tiny pieces and add them up:

$$dW = F\, ds \qquad W = \int F\, ds$$

2. The forces on the block

- **Weight** mg: straight down.
- **Normal force** N: the dome pushes back, at right angles to the surface, out along the radius.
- **Friction** $f = \mu N$: fights the motion, so it points down the slope.
- **Pull** T: the string lies along the surface, so it pulls up the slope.

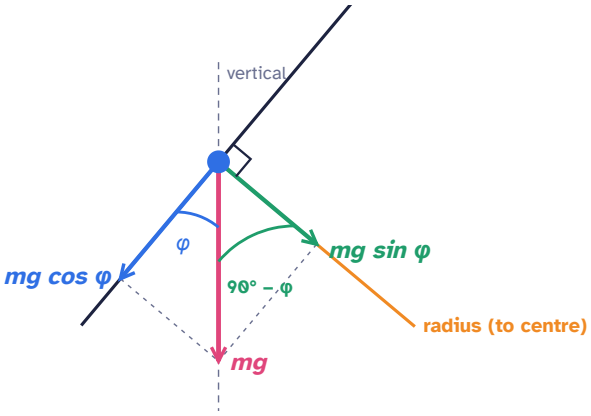
Remember: slowly means no acceleration. So the forces balance in every direction, and the kinetic energy never changes ($\Delta KE = 0$).

3. Where is the block? The angle φ

φ is the angle between the radius (centre → block) and the ground: 0 on the ground, $90^\circ = \frac{\pi}{2}$ at the top. A tiny turn $d\varphi$ moves the block along the curve by

$$ds = R\,d\varphi \qquad (\text{arc} = \text{radius} \times \text{angle, in radians})$$

4. Splitting the weight (the step most people get wrong)



1. The radius and the surface always meet at a **right angle**.
2. The radius makes φ with the ground. Turn both lines by 90°: radius → surface, ground → vertical. The angle stays the same.
3. So mg makes φ with the surface, and $90^\circ - \varphi$ with the radius.

along the surface: $mg\cos\varphi$ (down the slope)

into the dome: $mg\cos(90^\circ - \varphi) = mg\sin\varphi$

Check the ends: $\varphi = 0$: the surface is a wall, all of mg runs along it. $\varphi = 90^\circ$: the surface is flat, all of mg presses in. ✓

5. Balance the forces

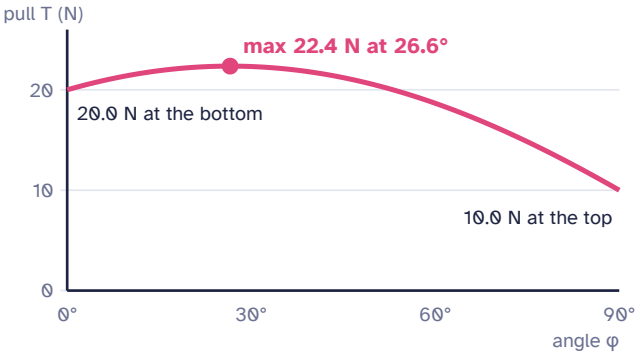
Along the radius (out = in). T and f lie along the surface, so they don't count:

$$N = mg\sin\varphi \quad \Rightarrow \quad f = \mu N = \mu mg\sin\varphi$$

Along the surface (up = down). N is across the surface, so it doesn't count:

$$T = mg\cos\varphi + \mu mg\sin\varphi$$

Careful: N is **not** mg. It changes along the dome, so friction changes too.



The pull for the worked example (2 kg, μ = 0.5). It is not largest at the bottom! (See P5.)

6. Add up the work

$$dW = T\,ds = (mg\cos\varphi + \mu mg\sin\varphi)\,R\,d\varphi$$

$$W = mgR\int_0^{\pi/2}\cos\varphi\,d\varphi \;+\; \mu mgR\int_0^{\pi/2}\sin\varphi\,d\varphi$$

$$= mgR\big[\sin\varphi\big]_0^{\pi/2} + \mu mgR\big[-\cos\varphi\big]_0^{\pi/2} = mgR(1) + \mu mgR(1)$$

$$\boxed{W_{\text{person}} = mgR\,(1+\mu)}$$

7. What the answer means

- mgR = weight × height: the work to lift the block by R. The shape of the path doesn't matter.
- μmgR = work against friction. This R is the **horizontal** distance, not the curved length.
- Smooth dome ($\mu = 0$): $W = mgR$. Makes sense.

Handy rule: when you drag something **slowly** over any curved surface, friction's work is $\mu mg \times$ horizontal distance.

8. Check it with energy (no integral)

$$W_{\text{total}} = \Delta KE = 0 \;\; (\text{slowly})$$

Normal force	always at right angles to the motion	$W_N = 0$
Gravity	block rises by R	$W_g = -mgR$
Friction	horizontal-distance rule	$W_f = -\mu mgR$
Person	the rest must cancel these	$W = mgR(1+\mu)$ ✓

9. With real numbers

$m = 2\ \text{kg},\ R = 1\ \text{m},\ \mu = 0.5,\ g = 10\ \text{m/s}^2$

Lifting	mgR	**20.0 J**
Friction	μmgR	**10.0 J**
Total	$mgR(1+\mu)$	**30.0 J**
Step-by-step simulation	adds up $T\,ds$ in 20,000 steps	**30.0 J** ✓

Common traps

Mistake	Answer you'd get	Why it's wrong
Forgetting friction	mgR = 20.0 J (too small)	The surface is rough.
Taking $N = mg$ everywhere	$mgR + \mu mg\frac{\pi R}{2}$ = 35.7 J (too big)	$N = mg\sin\varphi$ changes along the dome.
Thinking the pull is largest at the bottom	—	It peaks where $\tan\varphi = \mu$ (see P5).

Practice (try first, answers below)

1. Same dome ($m = 2$ kg, $R = 1$ m), but **smooth**. Find the work done by the person.
2. $m = 1$ kg, $R = 0.5$ m, $\mu = 0.3$, $g = 10$. Find the work from the ground to the top.
3. Worked example (2 kg, 1 m, $\mu = 0.5$), but the person stops at $\varphi = 60^\circ$. Find the work. *Hint: integrate from 0 to $\pi/3$.*
4. Worked example: find N and the friction at $\varphi = 30^\circ$.
5. Worked example: at what angle is the pull largest, and how large is it?

Answers

1. $\mu = 0 \Rightarrow W = mgR = 2 \times 10 \times 1 =$ **20.0 J**
2. $W = mgR(1+\mu) = 1 \times 10 \times 0.5 \times 1.3 =$ **6.50 J**
3. $W = mgR\sin\varphi + \mu mgR(1-\cos\varphi) = 20(0.866) + 10(0.5) =$ 17.3 + 5.00 = **22.3 J**
4. $N = mg\sin 30^\circ =$ **10.0 N**, $f = \mu N =$ **5.00 N**
5. $T = mg(\cos\varphi + \mu\sin\varphi)$ is largest when $\tan\varphi = \mu = 0.5$, i.e. $\varphi \approx$ **26.6°**. Then $T = mg\sqrt{1+\mu^2} =$ **22.4 N**, more than the 20.0 N at the bottom.

Original notes by Adithya Vikram. Every number here is computed by code and checked a second way (a step-by-step simulation or a numerical integral) before it is printed.